

Examining spatial patterns of selection and use for an altered predator guild

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Abstract Anthropogenic disturbances have altered species' distributions potentially impacting interspecific interactions. Interference competition is when one species denies a competing species access to a resource. One mechanism of interference competition is aggression, which can result in altered space-use of a subordinate species due to the threat of harm, otherwise known as a 'landscape of fear'. Alternatively, subordinates might outcompete dominant species in resource-poor environments via a superior ability to extract resources. Our goal was to evaluate spatial predictions of the 'landscape of fear' hypothesis for a carnivore guild in

Newfoundland, Canada, where coyotes recently immigrated. Native Newfoundland carnivores include red foxes, Canada lynx, and black bears. We predicted foxes and lynx would avoid coyotes because of their larger size and similar dietary niches. We used scat-detecting dogs and genetic techniques to locate and identify predator scats. We then built resource selection functions and tested for avoidance by incorporating predicted values of selection for the alternative species into the best supported models of each species. We found multiple negative relationships, but notably did not find avoidance by foxes of areas selected by coyotes. While we did find that lynx avoided coyotes, we also found a reciprocal relationship. The observed patterns suggest spatial partitioning and not coyote avoidance, although avoidance could still be occurring at different spatial or temporal scales. Furthermore, Newfoundland's harsh climate and poor soils may swing the pendulum of interspecific interactions from interference competition to exploitative competition, where subordinates outcompete dominant competitors through a superior ability to extract resources.

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Introduction

Natural and anthropogenic influences alter species distributions by reducing, expanding, or shifting a species' geographic range. These alterations may have a short- or long-term legacy (Brown et al. 2013) regardless of whether they are initiated as a result of a natural process, such as variability in weather patterns (Mehlman 1997), or anthropogenic activities, such as harvest (Ermala 2003) or species

introductions (Fraser et al. 2015). Ecologists are often concerned with the species that is undergoing a change in its distribution, particularly if the geographic range is reduced or fragmented (McKelvey et al. 2011). However, perhaps of equal concern are the subsequent direct (e.g., predation, Fordham et al. 2008; competition, Peers et al. 2013; disease transmission, Rochlin et al. 2013) and indirect (e.g., trophic cascades, Levi et al. 2012; apparent competition, Latham et al. 2013) impacts of a species, whose geographic range has expanded into a new ecological community.

Interspecific interference competition is when one species denies a competing species access to a resource (Lourenço et al. 2014). One means of denial is aggression (Case and Gilpin 1974), which in its most extreme forms can manifest as interspecific competitive killing, where a species kills a competitor, but not does consume the carcass, or intraguild predation, where the competitor is killed and consumed (Lourenço et al. 2014). Consequences of interspecific aggression can include numeric reductions in a subordinate predator via direct killing, or altered behavior (spatial or temporal avoidance) of a subordinate due to the risk of aggression or death (Berger 2010), potentially eliminating suitable habitat and creating a ‘landscape of fear’ (Laundré et al. 2001). For example, black-footed ferrets (*Mustela nigripes*) exhibited spatial avoidance of areas with high amounts of use from a dominant carnivore, the American badger (*Taxidea taxus*, Grassel et al. 2015). African wild dogs (*Lycaon pictus*) and cheetahs (*Acinonyx jubatus*) displayed spatial avoidance of lions (*Panthera leo*) and spotted hyenas (*Crocuta crocuta*), which was suggested to be an important mechanism for promoting their persistence (Creel and Creel 1996; Durant 2000).

Alternatively, species distributions may be driven by interspecific exploitative competition, where one predator outcompetes another via a superior ability to extract resources from specific habitats (Holt and Polis 1997). A superior ability to exploit resources in resource-poor habitats was theorized to explain higher kit fox (*Vulpes macrotis*) occupancy in comparison with dominant coyotes (*Canis latrans*) in New Mexico (Robinson et al. 2014). Indeed, the interplay between interference and exploitative competition is theorized to promote species persistence across heterogeneous landscapes with dominant predators excluding subordinate predators in more productive areas and subordinate predators outcompeting dominant predators in resource-poor habitats (Holt and Polis 1997; Verdy and Amarasekare 2010).

Although coyotes coexist with other predators within their historical geographic range, their expansion into new areas (Harrison 1986; Parker 1995) has led to impactful ecological consequences. Suppression of wolves, as well as landscape changes and increases in white-tailed deer (*Odocoileus virginianus*) densities facilitated coyote

dispersal across North America during the 20th century (Harrison 1986; Parker 1995). Climate change might also have played a role in coyote expansion by reducing snow depths and shifting deer distributions, thus allowing coyotes to occupy previously inhospitable northern and mountainous habitats (Dawe et al. 2014). Even though white-tailed deer populations may have aided coyote expansion (Harrison 1986; Parker 1995), deer are now regulated by coyotes in some areas via high fawn predation (Parker 1995). Similarly, coyote predation on calves has limited the recovery of woodland caribou (*Rangifer tarandus*) in southeastern Quebec (Crete and Desrosiers 1995). Moreover, the influence of coyotes is not limited to prey species. Several studies demonstrated changes in the abundance or distribution of mesopredator populations following coyote colonization. In Maine, red foxes (*Vulpes vulpes*) were only found at the home range boundaries between coyote family groups, thereby limiting fox habitat availability and carrying capacity (Major and Sherburne 1987; Harrison et al. 1989). Another study in Maine found changes in bobcat (*Lynx rufus*) diet and abundance following coyote establishment (Litvaitis and Harrison 1989).

The island of Newfoundland presents an ecosystem, where the recent arrival of coyotes may be impacting both prey and predator species. Coyotes colonized Newfoundland in the 1980s and have since spread across the island (Mahoney and Weir 2009) and become a major predator of caribou calves (Mumma et al. 2014; Bastille-Rousseau et al. 2016). This has garnered the attention of wildlife managers, because the Newfoundland caribou herd has declined by > 66% since 1998 (Weir et al. 2014), but remains the only extant population of woodland caribou not listed as endangered or threatened (COSEWIC 2002). Although the influence of coyotes on caribou is a focus of recent study (Bastille-Rousseau et al. 2016; Mahoney et al. 2016), there is no research exploring the spatial relationships between coyotes and native Newfoundland predators.

Our objective was to assess the spatial predictions under the ‘landscape of fear’ hypothesis. Specifically, we were interested in assessing spatial avoidance and niche overlap between coyotes and three Newfoundland carnivores: red foxes, Canada lynx (*Lynx canadensis*), and black bears (*Ursus americanus*). To evaluate potential spatial relationships, we used scat-detecting dogs and genetic techniques to locate and identify predator scats. These locations were then used in resource selection functions (RSF, Boyce et al. 2002) to identify habitat selection patterns for each species and assess spatial avoidance of coyotes by subordinate predators. We also compared differences in habitat use and spatial niche overlap between species to gain a more complete understanding of spatial niche partitioning by Newfoundland carnivores.

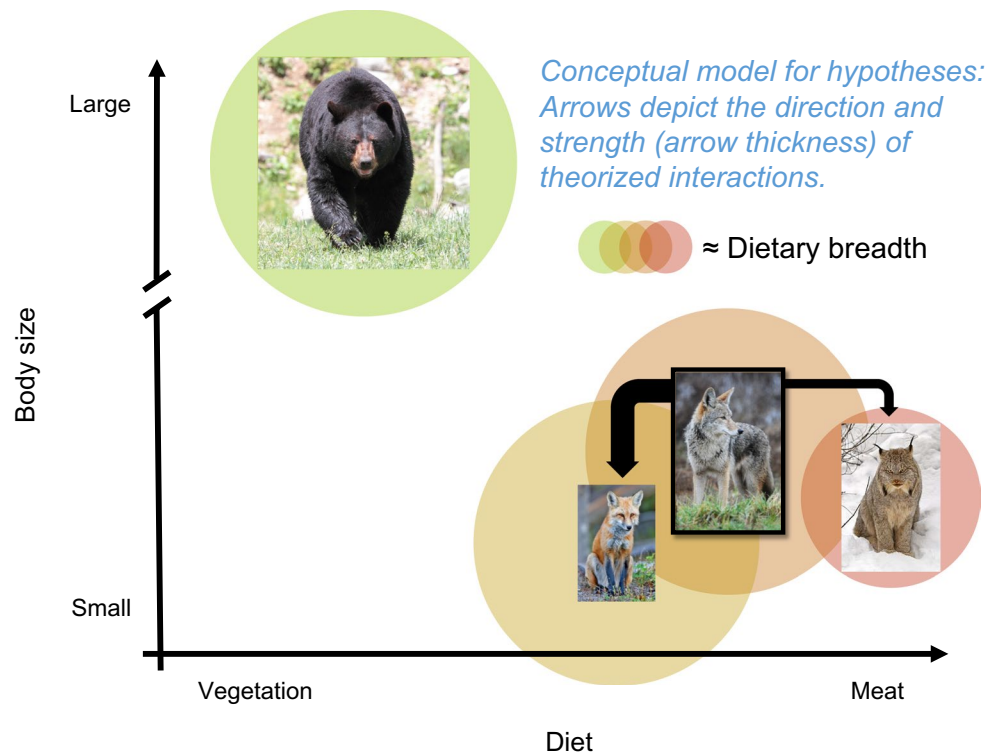


Fig. 1 Conceptual model of the hypothesized influence of coyotes on red foxes, Canada lynx, and black bears in Newfoundland, Canada. The direction and strength (indicated by arrows) of influence are dic-

tated by differences in size and overlap in diet composition (vegetation versus meat) and breadth (diversity of food items) between species

The Newfoundland system presents an interesting test case between three generalists (coyotes, red foxes, and black bears) and a specialist (Canada lynx) species. Furthermore, coyotes and foxes are adapted to search or chase prey over broad areas (cursorial hunters, Servin et al. 1991), while bears and lynx are suited for ambushing prey from close distances (Valkenburgh 1985). These differences in life history, along with differences in body size, permit us to theorize the spatial influence coyotes may have on native Newfoundland carnivores (Fig. 1). We hypothesized that coyote aggression, and thus, spatial avoidance would be a function of dietary niche overlap and body size differences (e.g., Palomares and Caro 1999; Fig. 1). We, therefore, predicted that foxes and lynx would both exhibit spatial avoidance of larger bodied coyotes, but that the strength of the effect would be stronger for red foxes because of a greater disparity in size and more similar dietary niche (Fig. 1). Avoidance of coyotes by foxes and lynx is consistent with reports of coyotes killing both these species (Sargeant and Allen 1989; O'Donoghue et al. 1995). Despite the comparatively large size of black bears, we did not anticipate finding any negative responses by the other carnivores, due to the wide separation in diets (Fig. 1). This allowed us to use bears as a pseudo-control and basis for comparison to the other species interactions. This work represents the first evaluation of the influence of coyotes on

the Newfoundland predators and contributes to our understanding of competition and spatial niche partitioning.

Materials and methods

Study site

The island of Newfoundland (111,390 km²) has a cool, maritime climate with interspersed coniferous forests, windswept barrens, and peatlands (McManus and Wood 1991). Historically, red foxes, Canada lynx, gray wolves, and black bears were the four large- to medium-sized, terrestrial, mammalian predators native to Newfoundland. Gray wolves were extirpated from Newfoundland prior to the 1930s (Lewis and Mahoney 2014), and although several wolves have been harvested in recent years, a breeding population has not re-established (Department of Environment and Conservation 2012). Coyotes likely colonized Newfoundland from mainland Canada by crossing sea ice (Lewis and Mahoney 2014).

Because of Newfoundland's declining caribou population, sampling for this study was originally designed to estimate predator densities on or adjacent to three caribou calving grounds (La Poile-LP, Middle Ridge-MR, and Northern Peninsula-NP) selected to capture the distribution

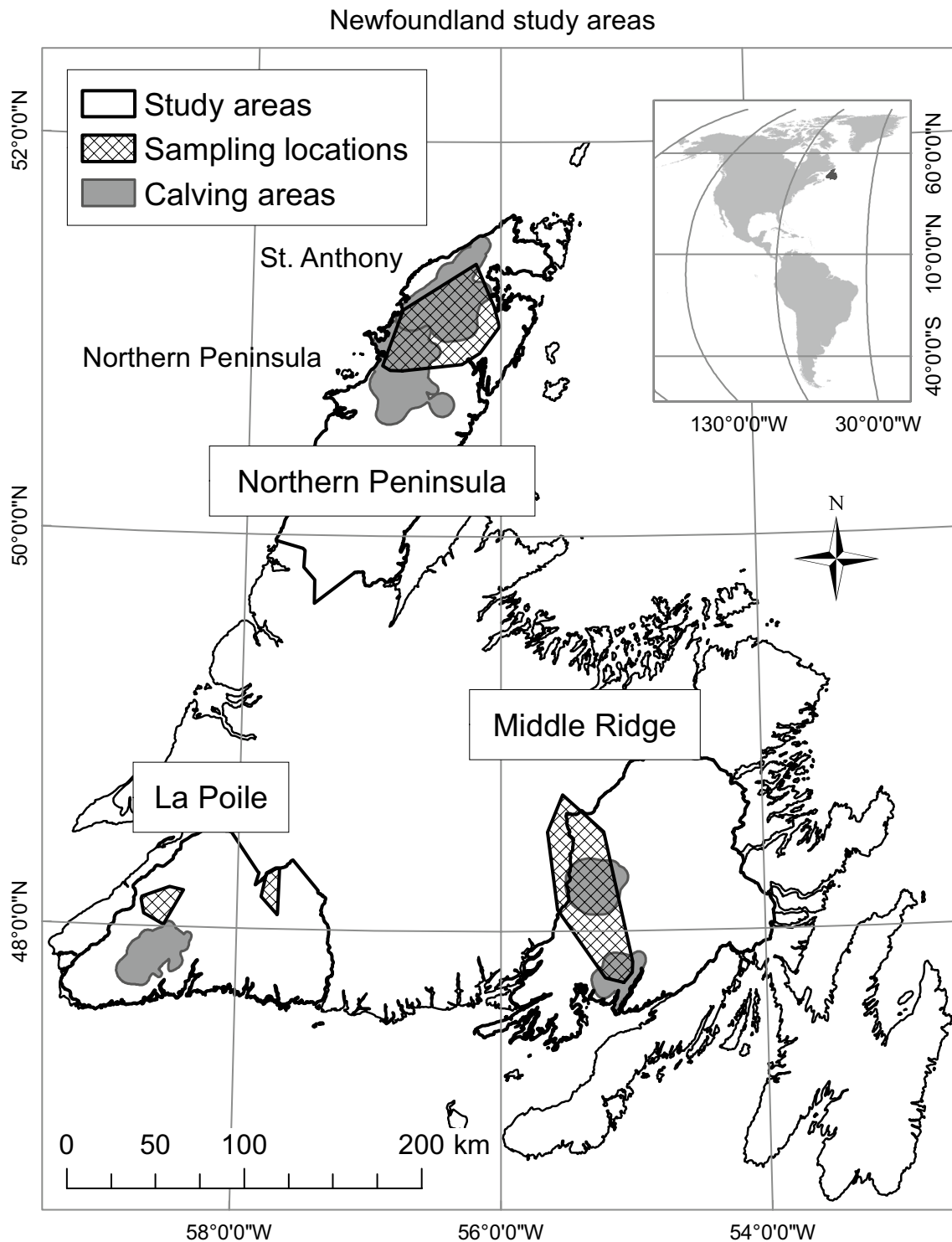


Fig. 2 Map of Newfoundland, Canada study sites, calving areas, and sampling locations

of Newfoundland's climatic variability. They were previously delineated using radio-telemetry data (Rayl et al. 2014) and correspond to four caribou herds (La Poile, Middle Ridge and the adjacent Northern Peninsula and Saint Anthony herds; Fig. 2).

Sampling and genetic methods

We collected red fox, Canada lynx, coyote, and black bear scats using scat-detecting dogs from 2009 to 2012 in Newfoundland. Grid cells (12×12 km) were overlaid across the

NP (16 cells) and MR (25 cells). Seventeen search locations were dispersed across the LP to match the sampling intensity of other study areas. Sampling occurred in June, July, and August of each year, but the LP was only sampled in 2009 because of logistical constraints (see Supplementary material for additional details). Scats were also collected along roads during travel to and from search locations. Although sampling commenced in June of each year, uncertainty remains regarding the date when scats were deposited; given scat degradation rates, which were likely exacerbated by Newfoundland's damp climate (Murphy et al. 2007), our effective sampling period likely began in May. A genetic sample was collected from several exterior locations on each scat (Stenglein et al. 2010) and stored in individual 2-ml tubes at room temperature (Frantzen et al. 1998) until processing (sampling methods, see Supplementary material for additional details).

All scat samples were extracted, including a negative control, using Qiagen¹ QIAamp DNA stool minikits (Qiagen Inc., Valencia, CA) in a laboratory reserved for the processing of samples (scat, hair, saliva, etc.) containing low quantities of DNA to limit the possibility of contamination. We identified scats to species using a mitochondrial DNA control region fragment analysis test (Mumma et al. 2014; Genetic methods, see Supplementary material for additional details).

Landscape covariates

We determined a suite of covariates theorized to be important predictors of red fox, Canada lynx, coyote, and black bear habitat selection patterns in Newfoundland. We used a land cover map (~ 80% accuracy) developed from a series of Landsat seven scenes with a 30 × 30 m resolution (Integrated Informatics Inc. 2014). We reclassified the 12 landcover classes into five habitat types: forest (deciduous, conifer, or mixed stands), coniferous scrub (stunted conifer forest), open (lichen and heathlands, barrens, rocky, burns, and non-vegetated surfaces), wetlands (fens and bogs), and water. We determined the habitat type present at each scat location and calculated the proportions of each habitat type at 100-, 250-, 500-, and 1000-m radii. Additional covariates calculated at each radii were elevation, habitat diversity (number of habitat types), and terrain roughness (standard deviation of slope, Grohmann et al. 2011). At each scale, we tested all covariates for collinearity (package caret, Kuhn et al. 2013).

Resource selection and spatial avoidance

To assess patterns of selection, we used scat locations and their associated covariates to build RSFs (Boyce et al. 2002) for each predator species under a use-available design. An RSF (w) of a vector of covariates ($\mathbf{x} = x_1, x_2, \dots, x_n$) is a function proportional to the probability of selection (Boyce et al. 2002). Given that our objective was to evaluate

$$\text{RSF} = w(\mathbf{x}) = \exp(\beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n) \quad (1)$$

fine-scale spatial relationships between species, we established availability by randomly sampling (1 location/km²) from within minimum convex polygons of scat locations from all four species within each sampling location (Fig. 2). We estimated coefficients for RSFs using mixed-effects logistic regression with study site as a random intercept in program R (package lme4, Bates et al. 2015; R Core Team 2014). Because LP had smaller sample sizes, we grouped the LP and NP study sites, whose landscapes were more similar than that of the MR (Fig. 2). Prior to building RSFs, all continuous covariates were placed on a 0–1 scale using a linear stretch (Johnson et al. 2004; linear stretch, see Supplementary material for additional details), thus enabling comparisons of the relative strength of coefficient values. We excluded water from our analysis to ensure independence of habitat covariates. We took a hierarchical approach to model selection and first built models to identify the most informative scale (point covariates, covariates at 100-, 250-, 500-, and 1000-m radii) for the analyses via Akaike's Information Criteria for small sample sizes (AICc, Burnham and Anderson 2002). Next, we tested (AICc, Burnham and Anderson 2002) linear and quadratic relationships for elevation for each species to determine if any non-linear responses were better supported. We then developed competing hypotheses (Table S3) a priori for each species and determined the best supported models via AICc (Burnham and Anderson 2002). We verified model fit through k-fold cross validation using 100 repetitions of fivefold cross validation with 10 bins of equal size (Boyce et al. 2002). We used our top models to predict a surface for each species of the relative probability of selecting a given location based on landscape covariates and set these values on a 0–1 scale using a linear stretch (Johnson et al. 2004; linear stretch, see Supplementary material for additional details). To test for spatial avoidance, we added the values from the predicted surfaces as potential risk covariates to the top models of the other predators. Using the package lmer (Zeileis and Hothorn 2002) in program R (R Core Team 2014), we conducted likelihood ratio tests to provide a more stringent evaluation of whether or not the top models for our predators were improved by the addition of an alternative predator's relative probability of selection.

¹ Any use of trade, firm, or product names is for descriptive purposes only and does not imply endorsement by the U.S. Government.

Table 1 Beta (β) coefficients and standard errors (SE) for each covariate (standardized between 0 and 1) of the best supported resource selection functions via Akaike Information Criteria for small sample sizes for red foxes, Canada lynx, coyotes, or black bears during spring and summer in Newfoundland, Canada

Covariate	Fox		Lynx		Coyote		Bear	
	β	SE	β	SE	β	SE	β	SE
Proportion forest	2.960	0.567	1.587	0.934	0.324	0.313	2.165	0.282
Proportion conifer scrub	2.191	0.595	2.843	0.943	0.156	0.340	0.854	0.310
Proportion open	5.887	0.596	5.183	0.997	5.662	0.309	5.097	0.319
Proportion wetland	2.324	0.575	− 0.873	1.127	− 0.228	0.327	0.932	0.294
Habitat diversity	3.211	0.454	2.365	0.796	2.969	0.306	3.934	0.274
Elevation	1.707	0.642	− 2.812	1.080	− 2.381	0.460	7.946	1.159
Elevation ²	NA	NA	NA	NA	NA	NA	− 10.168	1.797
Terrain roughness	− 0.040	0.336	− 2.642	1.041	0.523	0.220	0.587	0.176

All proportions were calculated at a 100-m radius. Elevation was the mean value within a 100-m radius. Terrain roughness was the standard deviation of slope within a 100-m radius. A quadratic term ($\beta \times \text{Elevation} + \beta \times \text{Elevation}^2$) was used for bears

Habitat use and niche overlap

To visually evaluate differences in habitat use between species, we plotted the distributions for each habitat type (excluding water) at the scale of analysis determined to be most informative for resource selection. To quantitatively assess the degree of spatial niche overlap between species, we conducted pairwise species' comparisons by calculating Spearman's correlation coefficients in program R (R Core Team 2014) using the relative probability of selection at each available location for each species.

Results

Similarities in selection were evident across species with a few notable differences. Covariates at a 100-m buffer were most informative (Table S1), and our full model (all covariates included) was best supported across all species (Table S3), although the best supported bear model included a quadratic term for elevation (Tables 1 and S2). Bears selected for intermediate elevations, foxes selected higher elevations, and lynx and coyotes selected lower elevations (Fig. 3). Foxes and lynx demonstrated avoidance of areas with higher terrain roughness, while coyotes and bears demonstrated selection (Table 1). All species demonstrated positive responses to habitat diversity and to the proportions of each habitat type, with the exceptions of negative responses to wetlands by lynx and coyotes (Table 1).

The majority of responses to the relative probability of selection by an alternative predator were significant and all were negative (Fig. 4a–d). Reciprocal responses between fox and lynx (Fig. 4a, b), lynx and coyote (Fig. 4b, c), lynx and bear (Fig. 4b, d), and coyote and bear were found (Fig. 4c, d). Although coyotes demonstrated avoidance of areas predicted to be selected by foxes (Fig. 4c), foxes did not avoid

areas predicted to be selected by coyotes (Fig. 4a), and no relationship was found between foxes and bears (Fig. 4a, d).

The distributions of habitat use and the estimated spatial niche overlap (Spearman's correlation coefficient) for each species pair further revealed differences in space-use between species. Lynx used higher proportions of conifer scrub and forest (Fig. 5a, b), while coyotes utilized a higher proportion of open habitats (Fig. 5d). Foxes used the least amount of open habitats and the highest proportion of wetlands (Fig. 5c, d). Bears also used higher proportions of wetlands and forests, intermediate levels of open habitats, and lower amounts of conifer scrub (Fig 5a–d). Foxes and bears demonstrated the highest level (0.921) of spatial niche overlap (Fig. 6). Intermediate niche overlap values were

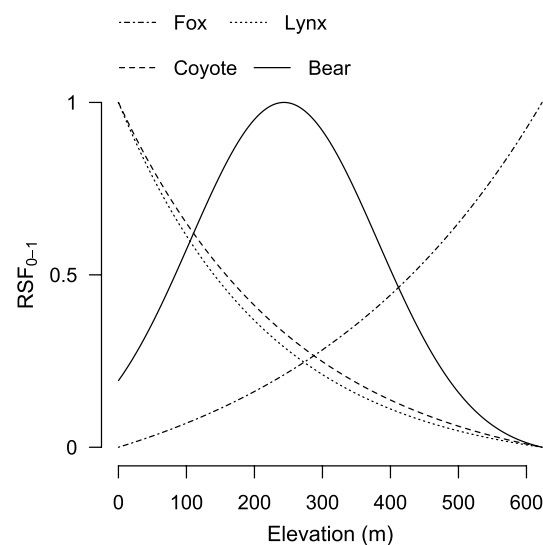


Fig. 3 Responses (set between 0 and 1) of red foxes, Canada lynx, coyotes, and black bears during spring and summer in Newfoundland, Canada to changes in elevation as predicted by the best supported resource selection functions (RSF) for each species

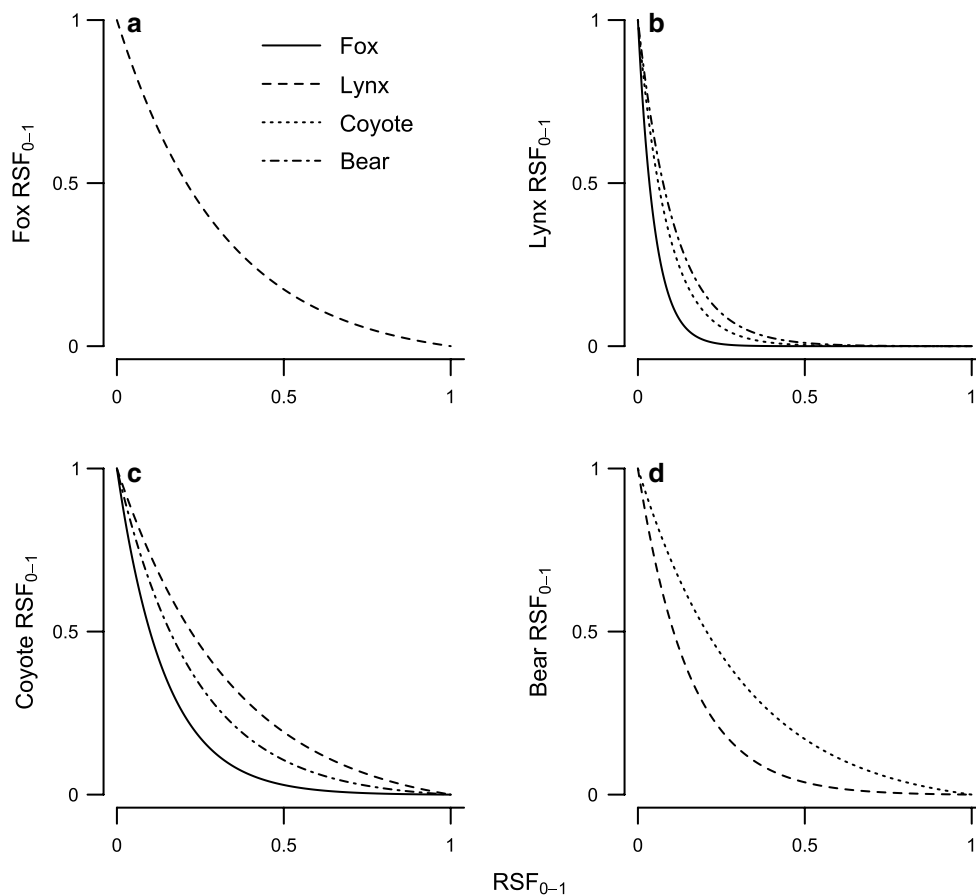


Fig. 4 Response (set between 0 and 1) of each predator species (red fox, Canada lynx, coyote, black bear) to the relative predicted probability of selection for each alternative species during spring and summer in Newfoundland, Canada. Resource selection function = RSF

observed for coyotes and foxes (0.772), coyotes and lynx (0.765), and coyotes and bears (0.782, Fig. 6). Lynx had comparatively low levels of spatial niche overlap with foxes (0.601) and bears (0.545, Fig. 6).

Discussion

In contrast to our spatial predictions of the ‘landscape of fear’ hypothesis, we did not see any evidence of red foxes avoiding areas selected by coyotes (Fig. 4a), and although we did see avoidance of coyotes by Canada lynx (Fig. 4b), the avoidance response was reciprocal (Fig. 4c). Furthermore, the large number of negative relationships between species (Fig. 4a–d), including negative relationships to our pseudo-control, bears (Fig. 4b, c), suggests that spatial niche partitioning is the most plausible explanation. Pairwise comparisons of spatial niche overlap reflect differences in space-use between species, and notably, we do not see any avoidance between foxes and bears, which show the highest levels of spatial niche overlap (Fig. 6).

Several factors may explain the seemingly absent (or inconsequential) spatial influence of coyotes on foxes and lynx. Species coexistence theory suggests that subordinate species are superior exploitative competitors and should predominate in resource-poor habitats (Holt and Polis 1997; Verdy and Amarasekare 2010). Newfoundland’s harsh climate and poor soils may swing the pendulum of interspecific interactions from interference competition to exploitative competition, where subordinates outcompete dominant competitors through a superior ability to extract resources. A similar trend was found for coyotes and kit foxes (*Vulpes macrotis*) in New Mexico, where the inability of coyotes to exploit prey-poor habitats was the primary driver of spatial segregation (Robinson et al. 2014). Differences in habitat productivity may, therefore, explain the contrast between our findings and those of other studies, which suggest that coyotes displace red foxes (Major and Sherburne 1987; Sargeant et al. 1987; Harrison et al. 1989). A similar explanation could apply between lynx and coyotes, although other studies have also failed to infer spatial interactions between coyotes and felids (*Felidae* spp). A study in California inferred coyote dominance over gray

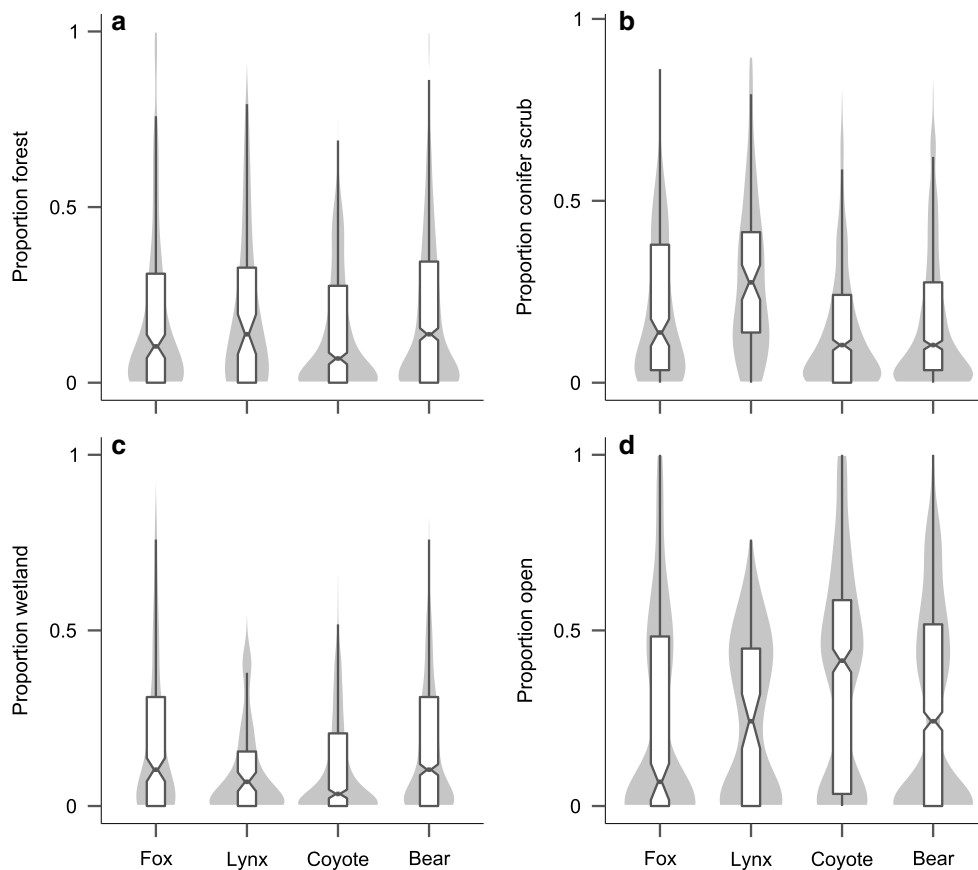


Fig. 5 Distributions (in gray) of habitat use (proportions of habitat within a 100-m radius) for red foxes, Canada lynx, coyotes, and black bears during spring and summer in Newfoundland, Canada. Boxplot

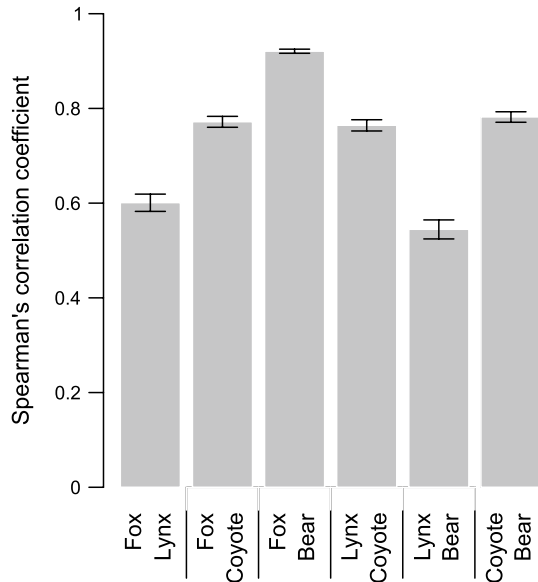


Fig. 6 Niche overlap (Spearman's correlation coefficient) and 95% confidence intervals for each pair of predator species (red fox, Canada lynx, coyote, and black bear) during spring and summer in Newfoundland, Canada

dimensions: median $- 1.58 \times \text{interquartile range (IQR)}/\text{number of observations } (n)^{1/2}$, 1st quartile, median, 3rd quartile, median $+ 1.58 \times \text{IQR}/n^{1/2}$

foxes (*Urocyon cinereoargenteus*), but not over bobcats and suggested that interactions between coyotes and bobcats may be rare (Fedriani et al. 2000).

Spatial and temporal aspects of the analyses are additional considerations. Our fine-scale analyses failed to detect red fox avoidance of locations with high probabilities of coyote selection. In contrast, avoidance was demonstrated in Maine when comparing red fox locations to coyote home ranges (Harrison et al. 1989). Alternatively, fine-scale daily movement patterns may allow subordinate carnivores to partition space and time with their dominant counterparts (Hayward and Slotow 2009; Kamler et al. 2013; Medina-Vogel et al. 2013), but our methodology did not allow us to evaluate fine-scale circadian patterns of selection. In addition, our spring and summer sampling may have missed spatial interactions during alternate seasons if habitat preferences and spatial niche overlap or the availability of resources change seasonally (Sinclair 1985; Jung et al. 2015). Indeed, coyotes during winter in the Yukon preferred dense spruce (*Picea* spp.) to avoid areas with deep snows (Murray et al. 1994).

Given the differences in hunting strategies between coyotes (cursorial) and lynx (ambush), spatial niche partitioning

between these two species is not surprising. Specialist lynx are adapted for ambushing snowshoe hares in habitats with ample cover (Squires et al. 2010; Ivan and Shenk 2016), while generalist coyotes prefer more open habitats (Thibault and Ouellet 2005; Boisjoly et al. 2010). These preferences align with our findings; lynx used habitats with greater cover (Fig. 5a, b), while coyotes used a greater proportion of open areas (Fig. 5d). More surprising were the differences in habitat use (Fig. 5a–d) and spatial niche overlap (Fig. 6) between foxes and coyotes, which were hypothesized to occupy a similar spatial niche due to their life history and generalist diets, although red foxes have been shown to select brushier habitats than coyotes (Theberge and Wedeles 1989).

Other considerations

Scat sampling permitted us to assess space-use for multiple species across a vast landscape without the expenses and risk associated with physical capture to researchers and animals. Many species of terrestrial carnivores use latrines for scat deposition (Gorman and Trowbridge 1989); therefore, scats may not represent an unbiased sample of animal locations. However, given that scats are frequently used as olfactory cues (Gorman and Trowbridge 1989), we contend that the use of scats to reveal spatial relationships between species is appropriate. Furthermore, we limited the impact of potential fine-scale selection patterns, unique to scat deposition, by testing covariates across multiple scales and ultimately using a 100-m radius for the determination of every covariate. Differences in the probability of detecting a scat, however, likely persisted between habitat types (Gutzwiller 1990), although we found scats in all terrestrial habitats. Furthermore, we assumed that any introduced sampling bias was reasonably consistent across species, thereby limiting the impact on our understanding of spatial relationships between species.

Conclusions

Inconsistent with the ‘landscape of fear’ hypothesis, we did not detect spatial avoidance of coyotes by red foxes or Canada lynx. The resource-poor environment of Newfoundland may provide foxes and lynx with a competitive advantage, which aligns with current theories of species coexistence (Holt and Polis 1997; Verdy and Amarasekare 2010). Furthermore, it suggests that coyotes may not be having a significant impact on native predator populations in Newfoundland, but we cannot rule out the possibility of coyote dominance at alternative spatial or temporal scales. In many systems, the role of competition and predation, in association with other factors, is poorly understood and the need for teasing apart these complex interactions and developing

more realistic models will only increase as global climate change continues to alter species’ geographic ranges with potentially community-destabilizing consequences.

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Author contribution statement MAM conceived and designed experiment, completed lab work, and wrote manuscript. JDH assisted in project conception and design and manuscript preparation. LPW provided conceptual input and editorial comments. NDR assisted with field work and provided editorial advice. CJZ conducted field work and provided knowledge of species niche breadth. TKF and JFO provided editorial comments. SPM facilitated funding and reviewed manuscript.

Compliance with ethical standards

Statement of human and animal rights Trained scat-detecting dogs were used to collect scats in this study and cared for in accordance with institutional and national guidelines. No animals or humans were used as research subjects in this article.

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